Element I: Testing Process

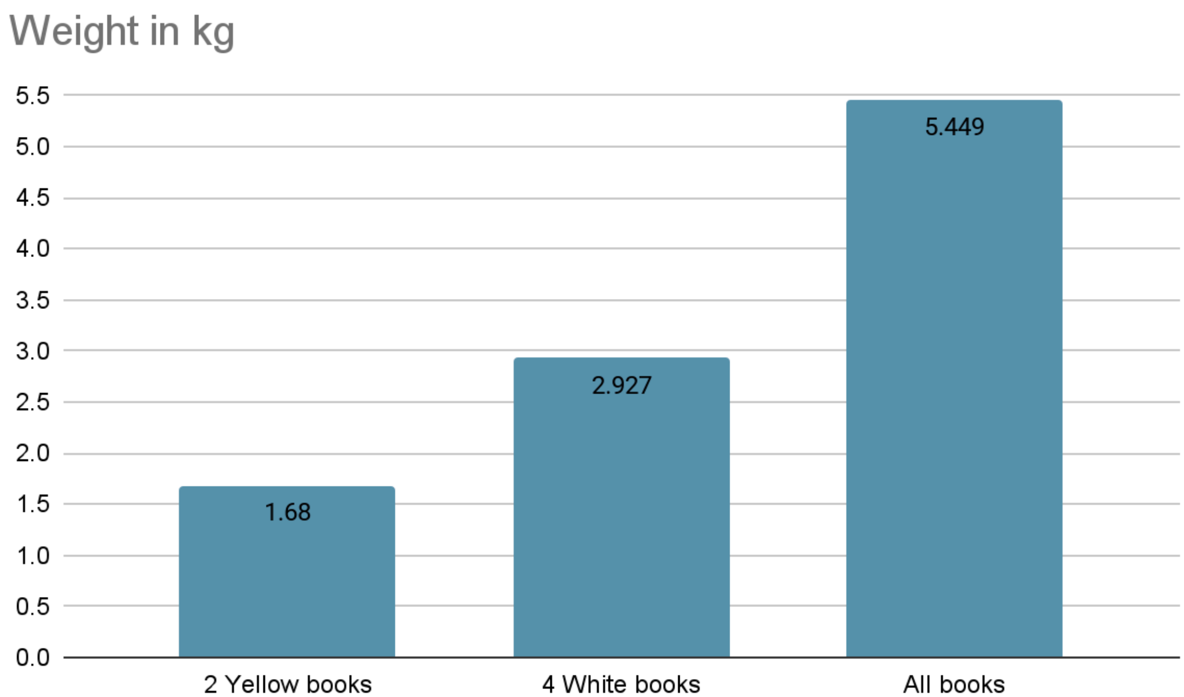
These were the procedures for test the prototype:

1. We first gathered and measured several books totalling to 5.449 kg
    a. We had 3 yellow books with a total of 2.522 kg and 4 white books that were 2.927 kg
2. We identified several controls and defining independent/dependent variables

    Experimental Controls: location of the test, the wall on the other side of book end,
    Independent Variable: weight of books
    Dependent Variable: ability of the book end to sustain weight

3. Place the book end on a flat surface to mimic conditions in the library
4. We tested first with 2 yellow books which were 1.68 kg by setting the book end near a wall to mimic the end of the usual bookshelf
5. The book end was able to sustain the weight so we tested another 4 trials so ensure it could still work out
6. We then wanted to increase the weight to see if the book end could still hold it so we decided to test with 4 white books which were 2.927 kg
7. We repeated this experiment for another 4 trials and it still worked without any damage
8. After it worked, we decided to use all of the available books which was a total of 5.449 kg
9. We repeated this another 4 trials to make sure it could still work out after several uses

We only changed one variable of the weight at a time but conducted 5 trials for each weight

Data Meaning and Summary of Implications

The data shows that the bookend is able to hold several kilograms of weight without breaking. It proves the stability of the bookend and that it would be successful for our stakeholders.

Each Implication from Summary of Data

Based on the test in Element H, the prototype was rated a 5.

Next Steps for Design/Process

The next steps for the design is to make it simpler. We went through many changes during the original construction. If we were to make a second bookend, the process should be intended to get the final design in a straightforward way, instead of the original roundabout way.